Program-A (CPU-Bound)

Time Quantum: 2ms
Involuntary Context Switches: 133
Voluntary Context Switches: 80000
Total Elapsed Time: 8.51s

Time Quantum: 5ms
Involuntary Context Switches: 54
Voluntary Context Switches: 80004
Total Elapsed Time: 9.20s

Time Quantum: 20ms
Involuntary Context Switches: 10
Voluntary Context Switches: 79997
Total Elapsed Time: 9.32s

Time Quantum: 100ms
Involuntary Context Switches: 1
Voluntary Context Switches: 80001
Total Elapsed Time: 7.94s

Time Quantum: 200ms
Involuntary Context Switches: 0
Voluntary Context Switches: 79996
Total Elapsed Time: 39.34s

Program-B (I/O-Bound)

Time Quantum: 2ms
Involuntary Context Switches: 147
Voluntary Context Switches: 79999
Total Elapsed Time: 11.87s

Time Quantum: 5ms
Involuntary Context Switches: 61
Voluntary Context Switches: 79998
Total Elapsed Time: 8.38s

Time Quantum: 20ms
Involuntary Context Switches: 7
Voluntary Context Switches: 80005
Total Elapsed Time: 8.98s

Time Quantum: 100ms
Involuntary Context Switches: 0
Voluntary Context Switches: 79973
Total Elapsed Time: 15.86s

Time Quantum: 200ms
Involuntary Context Switches: 0
Voluntary Context Switches: 80001
Total Elapsed Time: 17.79s

Cpu.c

```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>
#include <sys/resource.h>
#include <sys/time.h>

// CPU intensive work
void cpu_bound_task() {
    volatile long long x = 0;
    for(long long i = 0; i < 1e9; i++) {
        x += i;
    }
}

int main() {
    struct timeval start, end;
    struct rusage usage;

    gettimeofday(&start, NULL);

    // Create 8 child processes
    for(int i = 0; i < 8; i++) {
        if(fork() == 0) {
            cpu_bound_task();
            exit(0);
        }
    }

    // Parent waits
    for(int i = 0; i < 8; i++) {
        wait(NULL);
    }

    gettimeofday(&end, NULL);
    getrusage(RUSAGE_CHILDREN, &usage);

    double time = (end.tv_sec - start.tv_sec) +
                  (end.tv_usec - start.tv_usec) / 1e6;

    printf("CPU-BOUND RESULTS\n");
    printf("Involuntary Context Switches: %ld\n", usage.ru_nivcsw);
    printf("Voluntary Context Switches: %ld\n", usage.ru_nvcsw);
    printf("Total Elapsed Time: %.2f s\n", time);

    return 0;
}
```

IO.c

```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>
#include <sys/resource.h>
#include <sys/time.h>

// I/O simulation (sleep-based)
void io_bound_task() {
    for(int i = 0; i < 10000; i++) {
        usleep(100);  // simulate I/O wait
    }
}

int main() {
    struct timeval start, end;
    struct rusage usage;

    gettimeofday(&start, NULL);

    // Create 8 child processes
    for(int i = 0; i < 8; i++) {
        if(fork() == 0) {
            io_bound_task();
            exit(0);
        }
    }

    // Parent waits
    for(int i = 0; i < 8; i++) {
        wait(NULL);
    }

    gettimeofday(&end, NULL);
    getrusage(RUSAGE_CHILDREN, &usage);

    double time = (end.tv_sec - start.tv_sec) +
                  (end.tv_usec - start.tv_usec) / 1e6;

    printf("IO-BOUND RESULTS\n");
    printf("Involuntary Context Switches: %ld\n", usage.ru_nivcsw);
    printf("Voluntary Context Switches: %ld\n", usage.ru_nvcsw);
    printf("Total Elapsed Time: %.2f s\n", time);

    return 0;
}
```

```
sr@sr-VirtualBox: ~/OS
kernel.sched_rr_timeslice_ms = 2
sr@sr-VirtualBox:~/OS$ sudo taskset -c 0 chrt -r 20 ./cpu
Io-BOUND RESULTS
Involuntary CS= 133
voluntary CS= 80000
total time elapsed: 8.51 s
sr@sr-VirtualBox:~/OS$ sudo taskset -c 0 chrt -r 20 ./io
Io-BOUND RESULTS
Involuntary CS= 147
voluntary CS= 79999
total time elapsed: 11.87 s
sr@sr-VirtualBox:~/OS$ sudo sysctl -w kernel.sched_rr_timeslice_ms=5
kernel.sched_rr_timeslice_ms = 5
sr@sr-VirtualBox:~/OS$ sudo taskset -c 0 chrt -r 20 ./cpu
Io-BOUND RESULTS
Involuntary CS= 54
voluntary CS= 80004
total time elapsed: 9.20 s
sr@sr-VirtualBox:~/OS$ sudo taskset -c 0 chrt -r 20 ./io
Io-BOUND RESULTS
Involuntary CS= 61
voluntary CS= 79998
total time elapsed: 8.38 s
sr@sr-VirtualBox:~/OS$ sudo sysctl -w kernel.sched_rr_timeslice_ms=20
kernel.sched_rr_timeslice_ms = 20
sr@sr-VirtualBox:~/OS$ sudo taskset -c 0 chrt -r 20 ./cpu
Io-BOUND RESULTS
Involuntary CS= 10
voluntary CS= 79997
total time elapsed: 9.32 s
sr@sr-VirtualBox:~/OS$ sudo taskset -c 0 chrt -r 20 ./io
Io-BOUND RESULTS
Involuntary CS= 7
```

```
Io-BOUND RESULTS
Involuntary CS= 61
voluntary CS= 79998
total time elapsed: 8.38 s
sr@sr-VirtualBox:~/OS$ sudo sysctl -w kernel.sched_rr_timeslice_ms=20
kernel.sched_rr_timeslice_ms = 20
sr@sr-VirtualBox:~/OS$ sudo taskset -c 0 chrt -r 20 ./cpu
Io-BOUND RESULTS
Involuntary CS= 10
voluntary CS= 79997
total time elapsed: 9.32 s
sr@sr-VirtualBox:~/OS$ sudo taskset -c 0 chrt -r 20 ./io
Io-BOUND RESULTS
Involuntary CS= 7
voluntary CS= 80005
total time elapsed: 8.98 s
sr@sr-VirtualBox:~/OS$ sudo sysctl -w kernel.sched_rr_timeslice_ms=100
kernel.sched_rr_timeslice_ms = 100
sr@sr-VirtualBox:~/OS$ sudo taskset -c 0 chrt -r 20 ./cpu
Io-BOUND RESULTS
Involuntary CS= 1
voluntary CS= 80001
total time elapsed: 7.94 s
sr@sr-VirtualBox:~/OS$ sudo taskset -c 0 chrt -r 20 ./io
Io-BOUND RESULTS
Involuntary CS= 0
voluntary CS= 79973
total time elapsed: 15.86 s
sr@sr-VirtualBox:~/OS$ sudo sysctl -w kernel.sched_rr_timeslice_ms=200
kernel.sched_rr_timeslice_ms = 200
sr@sr-VirtualBox:~/OS$ sudo taskset -c 0 chrt -r 20 ./cpu
```

```
total time elapsed: 15.86 s
sr@sr-VirtualBox:~/OS$ sudo sysctl -w kernel.sched_rr_timeslice_ms=200
kernel.sched_rr_timeslice_ms = 200
sr@sr-VirtualBox:~/OS$ sudo taskset -c 0 chrt -r 20 ./cpu
Io-BOUND RESULTS
Involuntary CS= 0
voluntary CS= 79996
total time elapsed: 39.34 s
sr@sr-VirtualBox:~/OS$ sudo taskset -c 0 chrt -r 20 ./io
Io-BOUND RESULTS
Involuntary CS= 0
voluntary CS= 80001
total time elapsed: 17.79 s
sr@sr-VirtualBox:~/OS$
```